CyberLens

Even Realities G2 — Wearable SOC Assistant

Technical documentation for the CyberLens glasses interface. Covers architecture, voice commands, display constraints, ring navigation, module status, and privacy design principles.

Field	Value
Author	Luca Danisi
Version	1.0
Date	May 04, 2026
Hardware	Even Realities G2 Smart Glasses
Stack	TypeScript / Vite / Even Hub SDK / Node.js + Express
Portfolio	github.com/luckyluke1976/cybersecurity-portfolio

1. Overview

CyberLens Glasses is the wearable layer of the CyberLens SOC Assistant. It connects to a local Node.js server on port 3000 via the Even Hub SDK and surfaces SOC reference data directly in the analyst's field of view — no phone, no second screen required.

The project follows a strict local-first, privacy-first philosophy. Static lookups (ports, cheat sheets, subnets, playbooks, log codes) make zero external API calls. Claude AI is invoked only for CVE analysis and open-ended questions. External threat APIs are called only on explicit OSINT commands.

2. System Architecture

```
Voice Command / Ring Input
|
Even Hub SDK (TypeScript / Vite)
|
CyberLens SOC Server (Node.js / Express -- port 3000)
|
G2 Waveguide Display (56 chars/line, 12 lines max)
```

Server module categories:

Category	Modules	External calls

Static lookup	Port, Cheat Sheet, Subnet, Troubleshooting, IR Playbooks, Log Codes	None
AI-powered	CVE Lookup, open-ended fallback	Claude API
Threat intel	OSINT IP	ip-api.com, AbuseIPDB
Hash / File	Hash Lookup	VirusTotal API v3
Coaching	Tips — live interview coach	Soniox API (LIVE mode only)
Access control	Firewall / ACL Generator	None

3. Display Constraints

The G2 waveguide display is not a general-purpose screen. All server responses are formatted before transmission to the glasses. These limits are enforced server-side — not client-side.

Constraint	Value	Reason
Max line width	56 chars	Hard pixel limit on G2 bitmap font
Max lines/screen	10-12	Visible area of the waveguide
Character set	ASCII only	Unicode symbols render as '?' on the G2 font
Scroll method	Ring device	Thumb scroll navigates long output
Font style	Monospaced	Consistent column alignment in tables

4. Voice Commands

Trigger	Module	External API?
porta <number>	Port Lookup	No
cve <code>	CVE Details	Claude AI
subnet <ip/cidr>	Subnet Calculator	No
cheat <tool>	Cheat Sheet	No
trouble <scenario>	Troubleshooting	No
playbook <scenario>	IR Playbook	No
playbook <scenario> <phase>	IR Playbook phase	No
log? <code>	Log Code Reference	No
logref <category>	Log Code category	No
acl <rule>	ACL Generator	No
ip <address>	OSINT IP	ip-api.com + AbuseIPDB

5. IR Playbooks

12 incident response playbooks aligned to NIST SP 800-61. Each playbook covers all four NIST phases: Preparation, Detection & Analysis, Containment/Eradication/Recovery, and Post-Incident Activity. Each phase includes a curated list of commands relevant to that scenario.

Scenario	Scenario	Scenario	Scenario
Ransomware	Phishing	DDoS	Insider Threat
Data Breach	Malware	APT	BEC
Supply Chain	Cloud Incident	OT/ICS	Zero-Day

6. Ring Navigation

The companion ring device is the primary input method. Voice commands trigger queries; the ring navigates the response.

Ring Action	Result
Scroll	Navigate lines in current view
Single click	Select / confirm
Double-click	Go back / return to home screen

```
Home screen categories: TRIAGE | INVESTIGATE | RESPOND | REFERENCE
```

7. Tips Module — Live Interview Coach

The Tips module captures spoken conversation in real time and displays relevant IT/security concept definitions on the G2. Designed for technical interviews and certification study sessions.

Mode	Audio Source	STT Engine	Privacy
LIVE	G2 microphone	Soniox API (cloud)	Speaker diarization — your voice enrolled and filtered. Only interviewer tr
REMOTE	PC audio loopback	Faster-Whisper tiny int8 (local)	Fully local. Zero external calls.

Display behaviour:

Concepts stack newest-first. Ring scrolls. Auto-reset every 24h. Duplicates ignored (case-insensitive). Max 50 concepts. Each entry: concept name + definition, max 45 chars.

8. Module Status

Module	Status
Port Lookup	Working
CVE Lookup	Working

Subnet Calculator	Working
Cheat Sheet	Working
Troubleshooting Guide	Working
IR Playbooks (x12)	Working
Log Code Reference	Working
Firewall / ACL Generator	Working
OSINT IP Lookup	Working
Hash Lookup	Working
Home screen + ring navigation	Working
Tips — live interview coach	Built, testing pending
Wake word 'Veronica'	Planned
iPhone / LAN test page	Planned

9. Repository Structure

```
cyberlens-g2/
src/
main.ts # Entry point, Even Hub SDK init
display.ts # G2 formatter (56 char wrap, ASCII only)
commands.ts # Voice command parser + server router
vite.config.ts
package.json
.env # VITE_SERVER_URL=http://localhost:3000
```

```
cyberlens-soc-server/
server/
index.js # Express app, all module mounts
tips-router.js # Tips module router
tips-whisper.py # Faster-Whisper STT script (REMOTE mode)
data/
ports.json # 47 ports static lookup
cheatsheets.json # 10+ tool cheat sheets
mitre.json # Top 100 MITRE ATT&CK; techniques
timeline.json # Persisted incident timeline
playbooks/ # 12 IR playbook JSON files (NIST SP 800-61)
```

10. Privacy & Design Philosophy

Principle	Implementation
Local-first	All static lookups run with zero external calls
CIA Triad	Every design decision evaluated against Confidentiality, Integrity, Availability
Minimal API surface	Claude AI only for CVE + open questions. AbuseIPDB/VT only on explicit commands
No persistent logs	Session log in RAM, 24h TTL — nothing written to disk by default
Data minimisation	OSINT IP anonymises RFC1918 addresses before any external call

---

Built by Luca Danisi — aspiring SOC Analyst, Vienna. Finance & law background, pivoting into cybersecurity. Cert roadmap: CCST -> Security+ -> CySA+ -> ISO 27001 -> CISM
github.com/luckyluke1976/cybersecurity-portfolio